EMA51005 - Discrete Mathematics
UNIT III - Sets and Functions
Final Exam Answer Key

Answers are written in a simple university-exam flow: definition/given, required property, step-by-step working, and conclusion. Powers are typeset using normal superscripts.

Part A - Short Answers

1. **Define a relation on a set A.**
   Answer: A relation R on a set A is any subset of the Cartesian product $A \times A$. Thus,
   $$R \subseteq A \times A.$$
   If $(a, b) \in R$, we write aRb.

2. **Define symmetric difference of two sets.**
   Answer: The symmetric difference of A and B is the set of elements which belong to exactly one of the two sets.
   $$A \triangle B = (A - B) \cup (B - A).$$

3. **Define equivalence relation.**
   Answer: A relation R on a set A is called an equivalence relation if it is reflexive, symmetric and transitive.

4. **If $A = \{a, b, c\}$, find the power set $P(A)$.**
   Answer:
   $$P(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

5. **Define a characteristic function.**
   Answer: For $A \subseteq U$, the characteristic function of A is
   $$f_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

6. **State any two properties of the characteristic function.**
   Answer: Two important properties are
   $$f_{A \cap B}(x) = f_A(x) f_B(x),$$
   $$f_{A^c}(x) = 1 - f_A(x).$$

7. **Whether $R = \{(1, 1), (1, 2), (2, 1), (3, 4)\}$ is transitive on $A = \{1, 2, 3, 4\}$? Justify.**
   Answer: No. Since $(2, 1) \in R$ and $(1, 2) \in R$, transitivity requires $(2, 2) \in R$, but $(2, 2) \notin R$. Hence R is not transitive.

8. **Define hashing function.**
   Answer: A hashing function is a function which maps a key to an index or address in a hash table. It is commonly written as $h(x)$.

9. **Define permutation function with an example.**
   Answer: A permutation function is a one-to-one and onto function from a set onto itself. Example: for $A = \{1, 2, 3\}$,
   $$f(1) = 2, \quad f(2) = 3, \quad f(3) = 1.$$

10. **Write any two properties of relations.**
    Answer: Two properties are: (i) Reflexive: aRa for every $a \in A$; (ii) Symmetric: if aRb, then bRa.

Part B - 6 Mark Answers

1. **For any element $x \in A$, prove that $A \cup A = A$.**
   Answer: To prove two sets are equal, prove both inclusions.
   Let $x \in A \cup A$. Then $x \in A$ or $x \in A$. Hence $x \in A$. Therefore,

   $$A \cup A \subseteq A.$$

   Now let $x \in A$. Then certainly $x \in A \cup A$. Therefore,

   $$A \subseteq A \cup A.$$

   Hence,

   $$\boxed{A \cup A = A}.$$

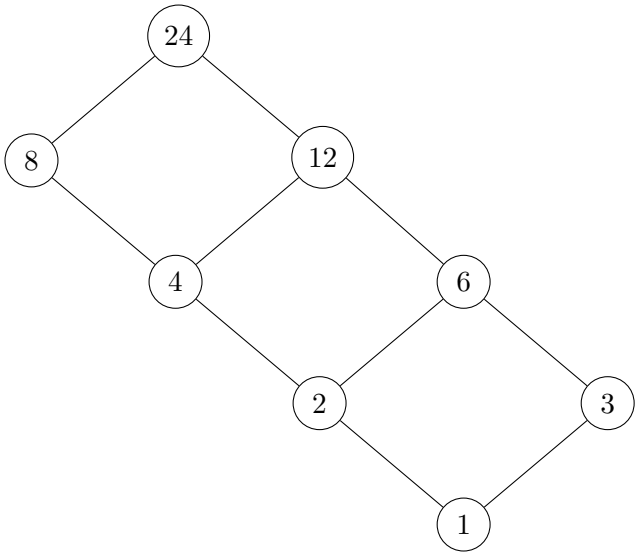
2. **Draw the Hasse diagram for D_{24}.**
   Answer: The positive divisors of 24 are

   $$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}.$$

   Under divisibility, the cover relations are

   $$(1,2), (1,3), (2,4), (2,6), (3,6), (4,8), (4,12), (6,12), (8,24), (12,24).$$

   A Hasse diagram is:



3. **Show that congruence modulo m is an equivalence relation on $\mathbb{Z}$.**
   Answer: Define aRb if and only if

   $$a \equiv b \pmod{m}.$$

   That means $m \mid (a - b)$.
   (i) Reflexive: For every $a \in \mathbb{Z}$,

   $$a - a = 0, \quad m \mid 0.$$

   Hence aRa.
   (ii) Symmetric: If aRb, then

   $$a - b = mk$$

   for some integer k. Therefore

   $$b - a = -mk = m(-k),$$

so bRa.

(iii) Transitive: If aRb and bRc, then

$$a - b = mk_1, \qquad b - c = mk_2.$$

Adding,

$$a - c = m(k_1 + k_2).$$

Hence aRc.

Therefore R is reflexive, symmetric and transitive. Thus R is an equivalence relation.

4. **Draw the Hasse diagram for the poset** $(\{1, 2, 3, 4, 5\}, \leq)$**.**
Answer: Since the usual order is a chain,

$$1 < 2 < 3 < 4 < 5.$$

Hence the Hasse diagram is



5. **If** A **and** B **are subsets of** U**, prove that** $f_{A\cap B}(x) = f_A(x)f_B(x)$**.**
Answer: Let $x \in U$. There are four possible cases.
If $x \in A$ and $x \in B$, then $f_A(x) = 1$ and $f_B(x) = 1$, so

$$f_A(x)f_B(x) = 1 = f_{A\cap B}(x).$$

If $x \notin A$ or $x \notin B$, then at least one of $f_A(x), f_B(x)$ is 0. Hence

$$f_A(x)f_B(x) = 0.$$

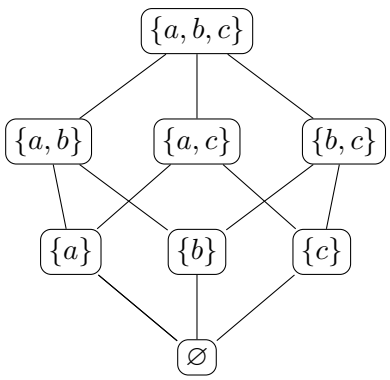
Also $x \notin A \cap B$, so $f_{A\cap B}(x) = 0$. Therefore in every case,

$$\boxed{f_{A\cap B}(x) = f_A(x)f_B(x)}.$$

6. **Draw the Hasse diagram of the relation contained on** $P(A)$ **where** $A = \{a, b, c\}$**.**
Answer:

$$P(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

Under set inclusion the Hasse diagram is:



7. **Using characteristic function prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.**
Answer: Let $x \in U$. For characteristic functions,

$$f_{A\cup B} = f_A + f_B - f_A f_B, \qquad f_{A\cap B} = f_A f_B.$$

Now,

$$\begin{aligned} f_{A\cup(B\cap C)} &= f_A + f_{B\cap C} - f_A f_{B\cap C} \\ &= f_A + f_B f_C - f_A f_B f_C. \end{aligned}$$

Also,

$$\begin{aligned} f_{(A\cup B)\cap(A\cup C)} &= f_{A\cup B} f_{A\cup C} \\ &= (f_A + f_B - f_A f_B)(f_A + f_C - f_A f_C). \end{aligned}$$

Since characteristic values are 0 or 1, $f_A^2 = f_A$. On simplifying,

$$f_{(A\cup B)\cap(A\cup C)} = f_A + f_B f_C - f_A f_B f_C.$$

Hence both characteristic functions are equal. Therefore,

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}.$$

8. **Prove that R is an equivalence relation if $(a, b) \in R$ iff $3a + 4b = 7n$ for some integer n.**
Answer: The condition means

$$3a + 4b \equiv 0 \pmod 7.$$

Since $4 \equiv -3 \pmod 7$,

$$3a - 3b \equiv 0 \pmod 7 \Rightarrow 3(a - b) \equiv 0 \pmod 7.$$

Since 3 is invertible modulo 7,

$$a \equiv b \pmod 7.$$

Thus this relation is exactly congruence modulo 7.
Reflexive: $a \equiv a \pmod 7$.
Symmetric: $a \equiv b \pmod 7 \Rightarrow b \equiv a \pmod 7$.
Transitive: $a \equiv b \pmod 7$ and $b \equiv c \pmod 7 \Rightarrow a \equiv c \pmod 7$.
Hence R is an equivalence relation.

9. **Show that R is an equivalence relation on positive integers if $(a, b) \in R$ iff ab is a perfect square.**
Answer: Write prime factorizations

$$a = \prod p_i^{\alpha_i}, \qquad b = \prod p_i^{\beta_i}.$$

The product ab is a perfect square iff every $\alpha_i + \beta_i$ is even.
Reflexive: $a \cdot a = a^2$ is a perfect square. Hence aRa.
Symmetric: If ab is a perfect square, then $ba = ab$ is also a perfect square. Hence $aRb \Rightarrow bRa$.
Transitive: Suppose ab and bc are perfect squares. Then for every prime p_i,

$$\alpha_i + \beta_i \equiv 0 \pmod 2, \qquad \beta_i + \gamma_i \equiv 0 \pmod 2.$$

Adding,

$$\alpha_i + \gamma_i \equiv 0 \pmod 2.$$

Hence ac is a perfect square. Therefore aRc.
Thus R is an equivalence relation.

10. **If $(a, b) \in R$ iff $a^2 + b$ is even, prove that R is an equivalence relation.**
Answer: For every integer a, a^2 has the same parity as a. Hence

$$a^2 + b \text{ is even } \iff a + b \text{ is even.}$$

So aRb iff a and b have the same parity.
Reflexive: $a^2 + a = a(a + 1)$ is even. Hence aRa.
Symmetric: If a and b have the same parity, then b and a have the same parity. Hence bRa.
Transitive: If a and b have the same parity and b and c have the same parity, then a and c have the same parity. Hence aRc.
Therefore R is an equivalence relation.

Part C - 10 Mark Answers

1. **Let $X = \{1, 2, 3, \ldots, 7\}$ and $R = \{(x, y) : |x - y|$ is divisible by $3\}$. Show that R is an equivalence relation and draw the graph of R.**
   Answer: The condition is equivalent to $x \equiv y \pmod 3$.
   Reflexive: $|x - x| = 0$, which is divisible by 3.
   Symmetric: $|x - y| = |y - x|$. Hence $xRy \Rightarrow yRx$.
   Transitive: If $3 \mid (x - y)$ and $3 \mid (y - z)$, then

   $$3 \mid [(x - y) + (y - z)] = x - z.$$

   Hence xRz. Thus R is an equivalence relation.
   The equivalence classes are

   $$[1] = \{1, 4, 7\}, \qquad [2] = \{2, 5\}, \qquad [3] = \{3, 6\}.$$

   Hence the graph consists of three complete directed components (including loops): one on $\{1, 4, 7\}$, one on $\{2, 5\}$ and one on $\{3, 6\}$.

2. **For ordered pairs of positive integers define $(a, b)R(c, d)$ whenever $ad = bc$. Show that R is an equivalence relation.**
   Answer: Reflexive: For (a, b),

   $$ab = ba,$$

   so $(a, b)R(a, b)$.
   Symmetric: If $(a, b)R(c, d)$, then

   $$ad = bc.$$

   Thus $cb = da$, so $(c, d)R(a, b)$.
   Transitive: Suppose $(a, b)R(c, d)$ and $(c, d)R(e, f)$. Then

   $$ad = bc, \qquad cf = de.$$

   Multiplying the first equation by f,

   $$adf = bcf.$$

   Using $cf = de$,

   $$adf = bde.$$

   Since $d > 0$, cancel d:

   $$af = be.$$

   Therefore $(a, b)R(e, f)$.
   Hence R is an equivalence relation.

3. **Define a relation on a set A, and list the properties of relations with an example.**
   Answer: A relation R on a set A is a subset of $A \times A$.
   Important properties are:
   Reflexive: $(a, a) \in R$ for every $a \in A$.
   Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$.
   Antisymmetric: $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$.
   Transitive: $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.
   Example: On $A = \{1, 2, 3\}$, the relation $\leq$ is reflexive, antisymmetric and transitive.

4. **On $A = \{1, 2, 3, 4, 5\}$, $(a, b) \in R$ if 3 divides $a - b$. Find R, R^{-1}, their domains/ranges, and the complement of R.**
   Answer: The congruence classes modulo 3 are

   $$\{1, 4\}, \quad \{2, 5\}, \quad \{3\}.$$

Hence

$$R = \{(1,1),(1,4),(4,1),(4,4),\\ (2,2),(2,5),(5,2),(5,5),(3,3)\}.$$

Since R is symmetric,

$$R^{-1} = R.$$

Therefore

$$\mathrm{Dom}(R) = \mathrm{Ran}(R) = A,$$

and similarly

$$\mathrm{Dom}(R^{-1}) = \mathrm{Ran}(R^{-1}) = A.$$

The complement $R^c = A \times A - R$ is

$$R^c = \{(1,2),(1,3),(1,5),(2,1),(2,3),(2,4),\\ (3,1),(3,2),(3,4),(3,5),(4,2),(4,3),(4,5),\\ (5,1),(5,3),(5,4)\}.$$

5. **Which of the given relations on $\{0,1,2,3\}$ are equivalence relations? State the properties lacking in the others.**
Answer: R_1**:** Contains only $(0,0),(1,1),(2,2),(3,3)$. It is reflexive, symmetric and transitive. Hence R_1 is an equivalence relation.
R_2**:** It is not reflexive because $(1,1)$ is missing. It is symmetric, but not transitive since $(0,2),(2,3) \in R_2$ while $(0,3) \notin R_2$. Hence not an equivalence relation.
R_3**:** Its classes are $\{0\},\{1,2\},\{3\}$. Hence it is reflexive, symmetric and transitive. So R_3 is an equivalence relation.
R_4**:** It is reflexive and symmetric, but not transitive because $(1,3),(3,2) \in R_4$ while $(1,2) \notin R_4$. Hence not an equivalence relation.
R_5**:** It is reflexive, but not symmetric because $(1,2) \in R_5$ while $(2,1) \notin R_5$. It is also not transitive; for example $(2,0),(0,1) \in R_5$ but $(2,1) \notin R_5$. Hence not an equivalence relation.

6. **Let $f(x) = x^2 - 2$ and $g(x) = x + 4$. Find the two compositions and the inverse functions if they exist.**
Answer:

$$f(g(x)) = f(x+4) = (x+4)^2 - 2 = x^2 + 8x + 14.$$
$$g(f(x)) = g(x^2 - 2) = x^2 + 2.$$

Therefore

$$f(g(x)) \neq g(f(x)).$$

For g,

$$y = x + 4 \Rightarrow x = y - 4,$$

so

$$\boxed{g^{-1}(x) = x - 4}.$$

For $f(x) = x^2 - 2$ on all of $\mathbb{R}$, f is not one-to-one because $f(x) = f(-x)$. Therefore f^{-1} does not exist as a function on all of $\mathbb{R}$.

7. **Using $h(x) = x^2 \pmod{11}$, insert the data $53, 13, 281, 743, 377, 20, 10, 796$ into cells 0 to 10.**
Answer: The hash values are

x	53	13	281	743	377	20	10	796
$h(x)$	4	4	3	3	9	4	1	5

Using the next higher unoccupied cell for a collision (linear probing), the placements are:

$$53 \to 4, \quad 13 \to 5, \quad 281 \to 3, \quad 743 \to 6, \quad 377 \to 9, \quad 20 \to 7, \quad 10 \to 1, \quad 796 \to 8.$$

Final table:

Index	0	1	2	3	4	5	6	7	8	9	10
Data	–	10	–	281	53	13	743	20	796	377	–

8. **For $h(x) = x \pmod{17}$, insert the given data using the next higher unoccupied cell.**
Answer: The data are $714, 631, 26, 373, 775, 906, 509, 2032, 42, 4, 136, 1028$.
The final placements are

Data	$h(x)$	Final cell
714	0	0
631	2	2
26	9	9
373	16	16
775	10	10
906	5	5
509	16	1
2032	9	11
42	8	8
4	4	4
136	0	3
1028	8	12

Thus the final hash table is

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Data	714	509	631	136	4	906	–	–	42	26	775	2032	1028	–	–	–	373

Note: the printed line "Where n is a positive integer, find $f(45)$" does not define a function f, so that isolated line cannot be evaluated from the question bank itself.

9. **Show that $f(x) = \dfrac{x}{x+4}$ is one-to-one and onto, and hence find the inverse.**
Answer: The printed mapping $f : \mathbb{R} \to \mathbb{R}$ is undefined at $x = -4$ and never takes the value 1. Hence the intended mapping must be

$$f : \mathbb{R} - \{-4\} \to \mathbb{R} - \{1\}.$$

One-to-one: Suppose

$$\frac{x_1}{x_1+4} = \frac{x_2}{x_2+4}.$$

Cross multiplying,

$$x_1(x_2+4) = x_2(x_1+4).$$

Therefore

$$4x_1 = 4x_2 \Rightarrow x_1 = x_2.$$

Hence f is one-to-one.
Onto: Let $y \in \mathbb{R} - \{1\}$. Set

$$y = \frac{x}{x+4}.$$

Then

$$yx + 4y = x \Rightarrow x(y-1) = -4y,$$

so

$$x = \frac{4y}{1-y}.$$

This belongs to the domain, hence every allowed y has a pre-image. Therefore f is onto.
Interchanging x and y gives

$$\boxed{f^{-1}(x) = \frac{4x}{1-x}}.$$

10. **Given relation matrices M_R and M_S, find matrices for $R \cup S$, $R \cap S$, $R \cdot S$, and $S \cdot R$.**
Answer:

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Using Boolean OR for union and Boolean AND for intersection,

$$M_{R\cup S} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix},$$

$$M_{R\cap S} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Using Boolean matrix multiplication,

$$M_{R\cdot S} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$M_{S\cdot R} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$